User Acceptance Testing Form



IT Ticket and Log System

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ITRI 624

Acceptance Form

PROJECT DETAILS

Project Name: *CSIS IT Ticket and Log System*

Project Manager: *Mej Selouna Visser*

Project Lead: *Shani Nezar & Lia Theunissen*

ACCEPTANCE DETAILS

Acceptance ID: *(N/A)*

Requested By: *CSIS IT Ticket and Log System Development Team*

Date Requested: *15 October 2026*

Description:

The CSIS IT Ticket and Log System is a web-based application for the NWU Computer Science and Information Systems department. It replaces manual email-based IT reporting with structured fault submission, unique ticket tracking, technician job cards and start/stop time recording. Students and staff can follow requests and exchange comments with technicians. The system provides automated NWU email notifications, escalation records for Central IT, immutable audit history and management reporting. This form records acceptance against the functional and technical specifications. Results, remedial actions and customer approval are completed after testing.

ACCEPTANCE CRITERIA

Criteria: *List the criteria against which the deliverable has been tested:*	*Functionality and Technical Usability* *Detailed criteria continue below.*

FUNCTIONAL REQUIREMENTS - F01-F28
F01 NWU email and password authenticate users and route each role to its dashboard using a secure session; invalid credentials show a clear error. Reporter entry follows approved D01.
F02 Names, NWU email and assigned roles persist correctly; only authorized administrators can change permissions.
F03 The structured submission form captures Title, Issue Description, Category, Urgency, Location and Station ID using predefined category and urgency choices, and confirms successful submission.
F04 Clear validation prevents incomplete or invalid tickets; corrected input is accepted. Record the approved institutional email domains.
F05 Each ticket has a unique Ticket ID, creation time, correct reporter association and initial state; all fields persist in MongoDB. See D04.
F06 Priority is saved accurately and organizes the queue so higher urgency work is visible first; hazardous priority is tested separately in T cases.
F07 Each reporter sees their own requests and status history; another reporter ticket is inaccessible through both UI and API.
F08 Authorized technicians see the permitted ticket queue, filter by status, inspect history and sort by priority. Cards show Station ID, Priority and time since submission. Labels follow D04.
F09 The ticket records the technician and claim time, moves to the approved active state and updates the dashboard and history.
F10 Only approved transitions succeed; the ticket, queue and audit history remain consistent for New/Pending, Assigned, In Progress, Escalated and Resolved as agreed.

F11 Start and stop timestamps persist, active work changes the state to In Progress, duration is calculated to the second and the displayed total matches MongoDB.
F12 The total adds both sessions correctly in minutes/hours, persists after reload and has timestamped audit entries for the time records.
F13 Claim-to-resolution elapsed time is calculated and saved separately from accumulated active timer time and creation-to-closure turnaround, using approved D06 definitions.
F14 An empty summary prevents closure; valid resolution records notes and closure time, archives the ticket from active queues and triggers the user notification.
F15 Status becomes Escalated to NWU IT with reason and traceable administrative hand-off; integration records referrals without requiring write access to central IT databases.
F16 Messages remain linked to the correct Ticket ID, in time order, with sender identity and timestamp; both permitted parties see the same persisted conversation.
F17 Each new public comment triggers an alert to the other party with the relevant ticket link and without exposing another ticket conversation.
F18 Authorized IT staff can read the note; the student/staff reporter cannot view it in the UI, API responses or notification contents.
F19 Posted communication remains immutable and cannot be deleted; the complete troubleshooting history persists under the approved retention approach D08.
F20 An automatic receipt contains the unique Ticket ID and submitted summary, reaches the correct NWU inbox and links to the ticket.
F21 The NWU SMTP service dispatches correctly addressed, template-based updates for all specified events with the right status, dynamic data and direct ticket link.
F22 Administrators filter historical tickets by date range, technician and hardware/category. Only matching records are returned and supply consistent inputs to reports.
F23 Admin graphs and tables match the source data, show workload distribution and recurring faults, and use correct time definitions including creation-to-closure turnaround.
F24 Both report formats contain the selected data and accurate metrics; files are readable and accessible only to the authorized role.
F25 Critical actions have actor and timestamp records, every state change exists in audit_logs and each resolved ticket identifies its technician and closure time; records resist modification.
F26 Ticket lists, status and public messages update asynchronously without a full page refresh, with near real-time visibility and recorded latency.
F27 Both user groups complete the full reporting and communication workflow successfully and receive the final resolution update.
F28 All job card, notification, audit and reporting records agree for the same ticket, with role boundaries maintained throughout.
TECHNICAL AND NON-FUNCTIONAL REQUIREMENTS - T01-T41
T01 MongoDB, Express and Node.js implement persistence and backend logic. React SPA in FS versus Vanilla JavaScript in TS is resolved and evidenced under D02.
T02 Node.js is v20.x or greater; Express.js v4.x baseline is verified; target is Ubuntu Server 22.04 LTS or configured Windows Server with a Node process manager such as PM2.
T03 The approved CSIS physical/virtual host supports the runtime and persistent ticket/audit storage; data location follows the agreed university boundary and D05.
T04 Functional parity is recorded for Chrome/Firefox/Edge, with FS Safari coverage; Windows 10/11, macOS and Linux browser access works without client installation.
T05 Mobile workflows match desktop functionality with readable responsive layouts and touch-friendly controls; FS Android 8.0+ and iOS 11.3+ support is documented or dispositioned.
T06 Core workflows remain usable on a 2 GB RAM dual-core desktop at 720p and a 2 GB quad-core mobile device with a 5-inch display, using keyboard/mouse or touch. Stable LAN, Wi-Fi or approved mobile access and 10/100 campus networking are supported.
T07 Purple/white branding and typography are consistent; deep purple #4B2C70 primary elements, slate secondary actions and labelled green/amber/red statuses follow the specified design.
T08 The specified WCAG 2.1 requirements, including contrast and keyboard access, are met; focus and labels support the workflows. Attach the audit scope and findings.
T09 Labels and errors are understandable; Log New Fault/recent-ticket access, station/priority/time details and lab/category analytics are present. Conceptual layouts are reviewed against approved designs.
T10 REST methods such as GET, POST and PUT exchange structured JSON between browser and backend; MongoDB storage and outbound Admin metrics follow the specified data flow.
T11 Web traffic uses HTTPS with TLS 1.2 or higher, protecting credentials and ticket data in transit; unencrypted access cannot expose protected data.
T12 Nodemailer sends notifications through the NWU gateway to @nwu.ac.za or @student.g.nwu.ac.za addresses. Port 587 with STARTTLS is the secure baseline; any port 25 use cited in TS p. 20 requires an approved encrypted relay configuration.
T13 Email processing does not block application requests and sustained test traffic leaves no unresolved notification backlog.

T14 Signed JWTs establish identity and role for protected requests; invalid or absent tokens cannot claim, close or alter tickets or audit data. D01 governs reporter entry.
T15 Management authentication operates using the internal module; no unavailable central Active Directory/SSO API is required for the initial release. Future integrations remain separately scoped.
T16 Reporters are limited to their own tickets; technicians manage approved workloads; reports/system logs and permission changes are restricted to authorized administrators. Resolve queue scope in D04.
T17 Stored passwords are salted bcrypt hashes with the configured high work factor; no plaintext credential storage is present. Record configuration evidence without copying secrets.
T18 Server validation using express-validator prevents NoSQL/injection attacks, XSS script execution and unauthorized database access, including when client validation is bypassed.
T19 Only the application server IP can connect to MongoDB; direct client or unapproved-host access is denied.
T20 Personal and operational records have documented encryption and secure handling controls; names and NWU emails support incident resolution. Attach POPIA and university policy review evidence for acceptance.
T21 Records are archived or anonymized after the academic cycle according to the approved policy, while audit traceability and immutable operational history are handled as agreed in D08.
T22 Standard input responses and ticket acknowledgement are within 2.0 seconds; record timings, network conditions and any failures. PP page-load checks are also captured.
T23 At least 50 concurrent connections are supported without data corruption or service decline. Capture response times, error counts and evidence that the notification queue drains.
T24 The system remains usable without client slowdown or significant lag. Record observed limits and agree a dataset and qualitative acceptance decision where no numeric budget is specified.
T25 Polling or supported WebSockets promptly refresh both views; one consistent technician assignment is retained and conflicting claims do not overwrite each other silently.
T26 Measure against the FS target of 99.9% during standard academic hours; record window, downtime and result separately from the technical target until D03 is approved.
T27 Measure both TS requirements: 98% uptime in that window and no more than 9.5 hours unplanned downtime per semester. Record period and resolve the differing targets under D03.
T28 All 10 students submit successfully without verbal help and each submission takes less than 60 seconds; attach participant timing evidence.
T29 Physical-risk tickets receive Critical/Hazardous priority and a prominent red labelled alert for immediate attention; no actual hazardous equipment is needed for testing.
T30 Technicians and management can identify repeated location-specific failures for investigation, including potential overheating/cooling patterns.
T31 Transactional handling restores the last known good state or a fully committed update; no partial or lost hazard ticket remains.
T32 At the predetermined interval the Lab Manager receives the escalation notification. Thirty minutes is an example in TS, not a fixed threshold until approved.
T33 Hardware-fault submission displays approved NWU laboratory safety reminders, including power-down guidance. Digital records show reported hazards and actions for institutional safety review and OHSA evidence.
T34 MongoDB backups run automatically and successfully at least once per 24-hour window; evidence identifies the latest recoverable backup.
T35 The database returns to its last known good backed-up state within 2 hours, with ticket, user, time-log and audit integrity verified.
T36 Every endpoint has JSDoc or Swagger documentation; ESLint or equivalent reports no critical style violations. snake_case data names and PascalCase modules follow TS conventions or documented exceptions.
T37 The maintainer can run the application in less than 20 minutes using the README; commented modular code supports future maintenance and deployment to other departments.
T38 A downloadable PDF quick-start guide, login-page FAQ and user/technician/Admin manuals cover access/login and password recovery, submission, categories, ticket status, timers, secure comments, reports, user roles and troubleshooting.
T39 Architecture, ERD, state transitions, wireframes, schemas, server deployment, SMTP setup and failure troubleshooting are available, current and professionally typed; external references follow NWU Harvard style.
T40 Source and technical documents are version controlled, secure SSH/HTTPS supports repository access and the documented update process delivers the approved release to the departmental server.
T41 The release uses approved free/open-source tooling and university resources with zero new licensing costs; initial launch is restricted to CSIS labs/offices within the 2026 scope.
SUPPORTING PROJECT PLAN REQUIREMENTS - P01-P06
P01 Automated API tests have a 100% pass rate and there are zero critical security vulnerabilities; attach the release-specific reports.
P02 The project-plan sub-2.0-second response target is assessed at the 100-user profile; retain this evidence separately from the TS minimum of 50 connections.
P03 The resolution email dispatches within 2 minutes of the trigger. Record inbox arrival separately so dispatch and delivery are distinguishable.

P04 The dedicated environment and email relay are provisioned and verified; unresolved dependencies have owners and cannot be recorded as passed without evidence.
P05 The CSIS Network Administrator receives repository ownership, VM/SSH and MongoDB administrative access through secure channels, environment/IP/connection documentation and user guides. Sponsor reviews deployment and handover evidence.
P06 A support owner, clear user guide and escalation hand-off exist; the planned automatic legacy-mailbox reply directs users to the portal and outstanding project issues have recorded actions.
OPTIONAL AND FUTURE REQUIREMENTS - O01-O04
O01 The design can be extended to screenshot/log uploads; implemented upload functionality is a future enhancement unless explicitly included in this release.
O02 Offline access is optional/future scope; record implementation evidence or an approved N/A disposition without making it a current mandatory gate.
O03 The optional printed output is readable through standard OS/browser printing; otherwise record approved N/A.
O04 Future Outlook/calendar and full central-system integration remain separately approved work; initial escalation record and local management login remain testable without them.
BASELINE DECISIONS BEFORE SIGN-OFF - D01-D09
D01 Confirm mandatory reporter login versus NWU-email submission without account creation.
D02 Confirm React MERN frontend versus the Vanilla JavaScript frontend stated in the technical specification.
D03 Approve the uptime target, operating calendar and measurement period; both source targets remain listed.
D04 Agree New/Pending and Assigned/In Progress transitions, escalation labels and technician queue permissions.
D05 Confirm internal hosting and the university data-location boundary versus the cloud option mentioned in the technical specification.
D06 Keep active work time, claim-to-resolution elapsed time and creation-to-closure turnaround distinct.
D07 Record both the 50-connection technical minimum and the 100-user project-plan performance profile.
D08 Agree how post-2026 archiving or anonymization preserves approved immutable audit history.
D09 Approve Critical/Hazardous priority, the hazard escalation timeout and Lab Manager recipient.

ACCEPTANCE RESULTS

Record a reviewer, date and result for each acceptance category below. Use linked evidence for individual checks.

- ***Testing methods used**:*
  _____
  _____
  _____
- ***Reviewer** who was formally responsible for undertaking the acceptance test (Owner and/or Super Users)*
  _____
  _____
  _____
- *Date the test was completed:_____.*
- **Overall result:** *Not assessed (complete after testing).*
  Total: _____ Passed: _____ Failed: _____ Blocked / Not run: _____

  ***Remedial Action Required**:*
  _____
  _____
  _____
  _____
  _____
  _____
  _____

Acceptance	Reviewer	Date	Result
Criteria 1: Functional Specification. *Refer to F01-F28 in the criteria register.*	▪ *Name of Reviewer:*	▪ *Date of Review:*	▪ *Result of Review (Failed, Passed, Exceeded Criteria + any remedial actions required):*
Criteria 2: Technical Specification. *Refer to T01-T41 in the criteria register.*	▪ *Name of Reviewer*	▪ *Date of Review*	▪ *Result of Review (Failed, Passed, Exceeded Criteria+ any remedial actions required):*
Criteria 3: Supporting Project Plan Requirements. *Refer to P01-P06 in the criteria register.*	▪ *Name of Reviewer*	▪ *Date of Review*	▪ *Result of Review (Failed, Passed, Exceeded Criteria+ any remedial actions required):*
Criteria 4: Optional and Future Requirements. *Refer to O01-O04; record approved N/A when not in release.*	▪ *Name of Reviewer*	▪ *Date of Review*	▪ *Result of Review (Failed, Passed, Exceeded Criteria+ any remedial actions required):*

SUPPORTING ACCEPTANCE RECORD

Project Team: Shani Nezar and Lia Theunissen
Project Sponsor: Selouna Visser
Release or Build: _____
Environment and URL: _____
Acceptance Decision: *Not assessed*
Evidence and defect record location: _____
Conditions and baseline decisions: Resolve D01-D09 before sign-off.
Record Pass, Fail, Blocked, Not run or approved N/A for each check in linked evidence. Record defect owners, remedial actions and retest outcomes before customer approval.

CUSTOMER APPROVAL

Supporting Documentation:
CSIS IT Ticket Log System Functional Specification v1.0.2
CSIS IT Ticket Log System Technical Specification v1.0.1
IT Ticket and Log System Project Plan, revision 1.2, and UAT evidence record

Signature: _____ **Date:** ___/___/____